Tidal potential measured on a coastal–inland optical-clock baseline

Abstract

The Moon and the Sun alter Earth's gravitational potential through their tidal fields and the responses of the solid Earth and oceans. Advances in optical clocks and fiber-optic networks have turned clocks into precision probes for probing spacetime, providing an experimental foundation for the long-term vision of using a global clock network to monitor Earth's gravitational potential. Changes in the tidal potential alter the relative ticking rates of clocks at different locations. Here, we use strontium and ytterbium clocks separated by 692 km to detect tidal variations in the potential difference between the coastal region of Shanghai and the inland region of Wuhan. Based on 545.9 hours of observations spanning 353 days, the agreement between the joint measurement of solid-Earth and ocean loading and model predictions is 0.96 ± 0.11. Using spectral analysis techniques, we separated the principal lunar tidal constituent M_2 and the principal solar tidal constituent S_2 at 22.361 and 23.147 μHz, respectively, with signal-to-noise ratios of 16.7 dB and 9.3 dB. This experiment demonstrates that optical clock networks can already be used for high-precision gravitational potential calibration and detection, laying a foundation for related applications.